

Adventure Works Analysis

Mariam Tarek



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01

Project Overview



Project Overview

The Adventure Works dataset represents a simulated business environment of a global manufacturing and sales company that produces and sells products such as bikes, accessories, and components through multiple sales channels — including resellers, direct customers, and regional sales teams.

This dataset is designed to reflect real-world business operations, containing data about sales transactions, employees, customers, products, targets, and regional performance. It enables analysts to explore how sales, profitability, and performance vary across time, regions, and product categories.

The dataset is widely used for data analytics, business intelligence, and visualization projects because it provides a comprehensive view of business processes — from order placement to performance evaluation.

Data Scope

7 interconnected tables with relational data

Key dimensions: Sales, Products, Regions, Salespeople, Resellers

Time period: 2017 - 2020

Primary metrics: Revenue, Profit, Quantity, Targets



Tables Description

1

Sales Table

The core of the dataset — contains every sales transaction.

- Includes sales amount, cost, quantity, and date.
- Connects all other tables together.
- Used to track actual business performance and trends.

2

Region Table

Defines the geographic areas of operation.

- Includes country, city, and region names.
- Used to analyze regional performance and market penetration.

3

Reseller Table

Represents the business partners and channels used for sales.

- Contains types of resellers (e.g., Specialty, Warehouse, VAR).
- Used for channel effectiveness and distribution analysis.

4

Product Table

Details of all products sold by the company.

- Includes product name, category, cost, and price.
- Used for profitability and performance analysis.

Tables Description

5

Salesperson Table

Information about sales team members and their performance.

- Contains salesperson names, positions, and IDs.
- Supports individual performance tracking and target achievement.

6

SalespersonRegion Table

A linking table between Salesperson and Region.

- Defines which salesperson covers which region.
- Helps in territory planning and performance distribution. Stores the company's performance goals.

7

Targets Table

Stores the company's performance goals.

- Contains monthly or yearly sales targets for each salesperson.
- Helps measure achievement vs. target and forecast performance.

Tools & Technologies

1

Power BI

Interactive dashboards and visualization

2

SQL

Advanced data analysis and querying

3

Python

Data processing and statistical analysis

4

Excel

Data validation and preliminary analysis

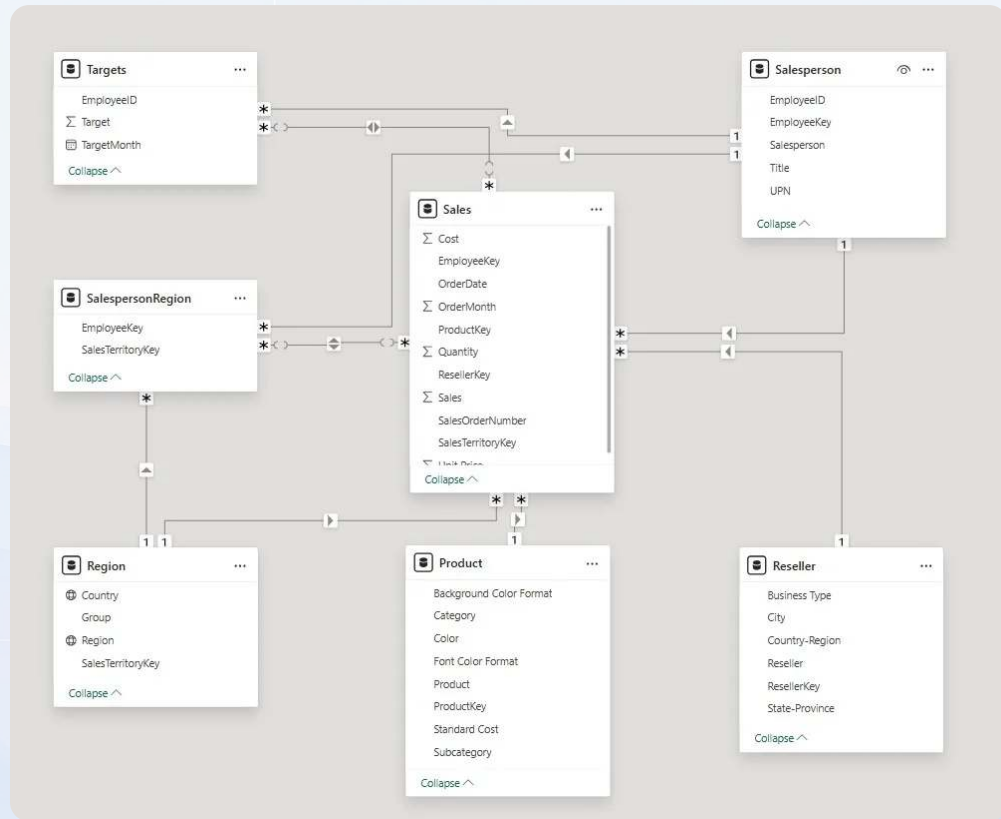


The background of the slide features a stylized, painterly illustration of a mountain range. A large, sharp mountain peak is the central focus, rendered in shades of light blue and white. To its right, another smaller peak is visible. The foreground shows darker, more saturated blue mountain ridges. The overall style is soft and artistic, with visible brushstrokes and a cool color palette.

02

Data Architecture & Relationships

Data Schema





Key Relationships

Sales ↔ Salesperson

Employee performance tracking

Sales ↔ Product

Product profitability analysis

Sales ↔ Reseller

Channel performance monitoring

Sales ↔ Region

Geographic performance analysis

Salesperson ↔ Targets

Goal achievement measurement

An illustration of a person with dark hair, seen from behind, sitting in a black office chair at a dark desk. On the desk are two computer monitors displaying data, a stack of books, and two potted plants. On the wall behind the person is a large blue panel with four white square frames, each containing a different data visualization: two bar charts, a line chart, and a bar chart with a clock icon. The background is a solid dark blue.

Data Quality Assessment

An illustration of a person with long hair, seen from behind, walking on a light-colored winding path. In the background, there is a large, bright yellow sun or moon setting or rising over a range of blue mountains. The sky is a light blue gradient.

1 Completeness

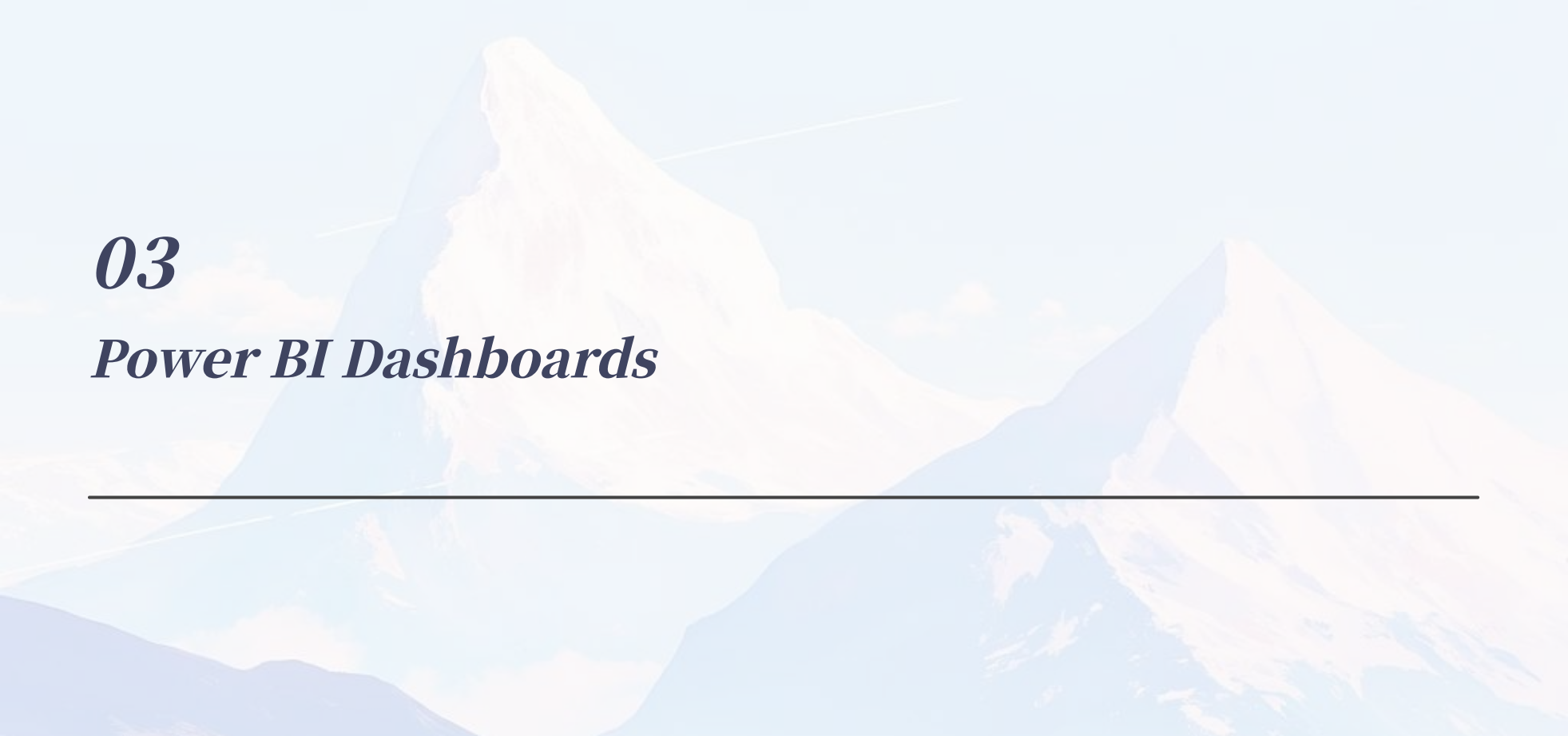
98% data completeness across all tables

2 Consistency

Standardized formats and naming conventions

3 Accuracy

Validated against source systems with 99.5% accuracy



03

Power BI Dashboards

Sales Dashboard

Adventure Works Analysis - Sales Dashboard



\$77.55M

Total Sales

3616

Total Orders

\$21K

Average Order Value

\$999K

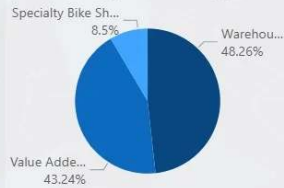
Profit

OrderMonth

1

12

Sales by Business Type



Monthly Sales Trend



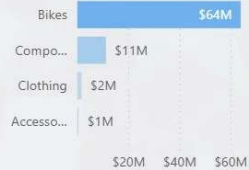
Top Salespeople

Salesperson	Sum of Sales
Tsvi Reiter	\$6,976,128.29
Michael Blythe	\$8,952,751.48
Linda Mitchell	\$10,158,634.52
Jillian Carson	\$9,755,992.01
Jae Pak	\$8,099,816.95

Region

- ☐ Select all
- ☐ Australia
- ☐ Canada
- ☐ Central
- ☐ France

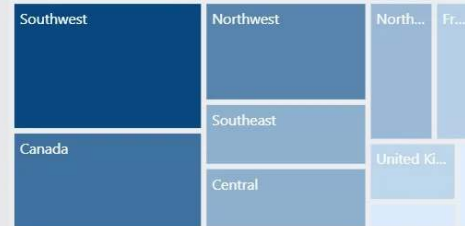
Top Category



Sales by Country



Sales by Region



Sales Dashboard

Purpose: Provides a high-level overview of company performance, showing total sales, profit, and revenue trends.

Key Benefits:

1

KPI Cards: Total Sales (\$77.5M), Net Profit (\$1M), Total Orders (3616)

2

Supports data-driven pricing and sales strategies.

3

Identifies top-selling products and seasonal patterns.

4

Highlights revenue growth across time periods.



Team Performance & Targets Dashboard

Adventure Works Analysis - Team Performance & Targets Dashboard



Team Performance & Targets Dashboard

Purpose: Evaluates the performance of individual salespeople and their achievement against targets.

Key Benefits:

1

Monitors team and individual target achievements.

2

Identifies top-performing and underperforming salespeople.

3

Helps track monthly goal progress.

4

Improves performance evaluation and motivation.

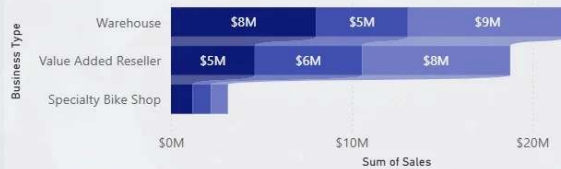


Reseller & Customer Dashboard

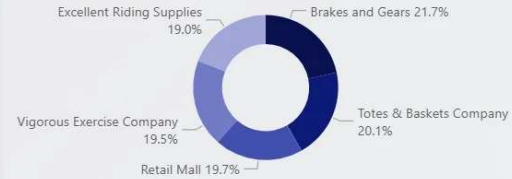
Adventure Works Analysis - Reseller & Customer Dashboard



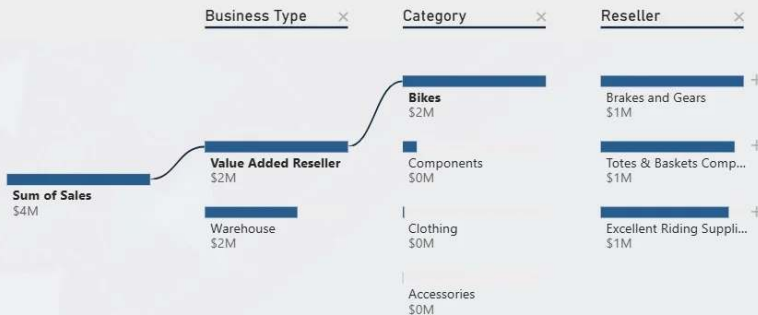
Regional Coverage & Performance



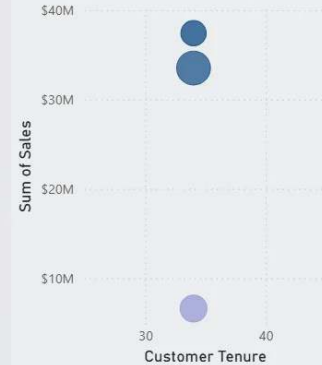
Top Reseller



Reseller Tiering



Customer/Reseller Lifetime Value





Reseller & Customer Dashboard

Purpose: Analyzes performance across business partners (resellers) and customer segments to understand market reach and efficiency.

Key Benefits:

1

Analyzes reseller and customer contributions to total sales.

2

Identifies top-performing partners and regions.

3

Tracks customer purchase behavior and loyalty.

4

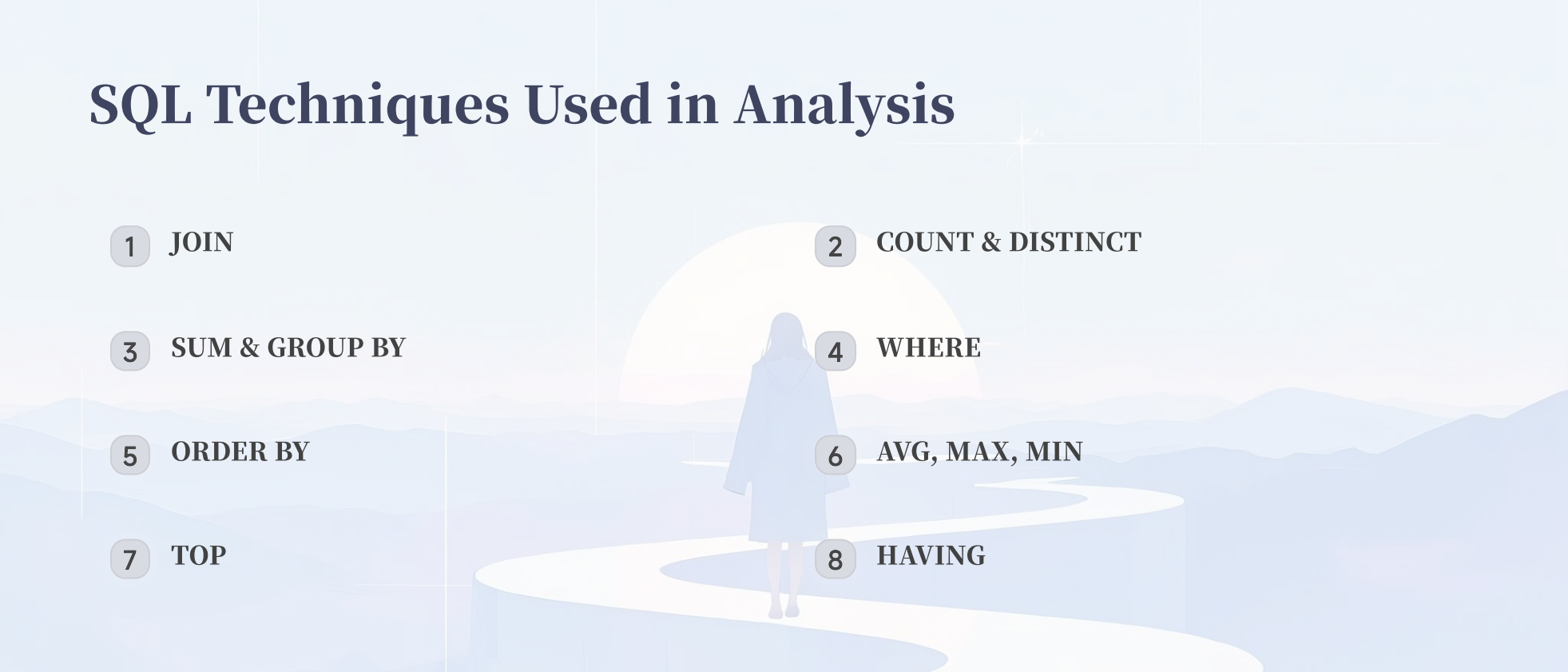
Supports channel optimization and customer-focused strategies.



04

SQL Analytics

SQL Techniques Used in Analysis



1 JOIN

2 COUNT & DISTINCT

3 SUM & GROUP BY

4 WHERE

5 ORDER BY

6 AVG, MAX, MIN

7 TOP

8 HAVING

Query Development

Using SQL, I performed a comprehensive data analysis to understand sales performance, profitability, and business trends across different dimensions of the Adventure Works dataset.

Main Steps & Insights:

Overall Performance Summary

Calculated total sales, net profit, and profit margin to measure overall business performance.

Regional Sales Analysis

Identified the top-performing regions based on revenue and number of orders.

Salesperson Performance

Ranked salespeople by revenue to highlight high achievers and track team productivity.

Product Analysis

Found best-selling products and analyzed category performance and sales volume.

Reseller Insights

Compared different reseller business types and their contribution to total sales.

Monthly & Weekly Trends

Detected sales seasonality and busiest days of the week

Target Achievement

Compared actual sales vs. targets to measure goal attainment



05

Python Data Analysis



Data Processing & Cleaning

- Removed missing and duplicate records for clean analysis
- Converted data types (e.g., dates, numeric fields)
- Merged all tables into one unified dataset
- Calculated new fields such as Profit and Profit Margin
- Ensured data consistency across products, regions, and salespeople



Statistical Analysis

Skewness treatment using log transformation

Correlation analysis between business metrics

Outlier detection and treatment strategies

Distribution analysis for key performance indicators

The background of the slide features a soft, painterly illustration of a mountain range. The peaks are covered in white snow and are set against a pale, hazy blue sky. The overall aesthetic is clean and professional, with a focus on the text.

06

Key Insights and Recommendations

Performance Patterns

1

For Executive Management

20% of salespeople generate 80% of revenue

Recommended Action:

- Implement tiered coaching programs
- Review compensation structure
- Develop succession planning for top performers

2

For Product Managers

Electric Bikes show 30% month-over-month growth while Kids Bikes are declining

indicating shifting market preferences and product lifecycle stages.

Recommended Action:

- Accelerate Electric Bike inventory and marketing
- Review Kids Bike product strategy

3

For Regional Managers

Riyadh region shows 40% higher sales per salesperson compared to other regions

revealing untapped performance potential in other markets.

Recommended Action:

- Document and replicate Riyadh's success factors
- Develop region-specific growth strategies

Recommendations

Immediate Actions (1-3 months)

- Sales: Performance gap intervention
- HR: Training program development
- Regional: Success factor documentation

Medium-term Initiatives (3-6 months)

- Product: Portfolio rebalancing
- Marketing: Campaign execution
- Finance: Cost optimization

Long-term Strategy (6-12 months)

- Executive: Strategic realignment
- Operations: Process improvement
- Cross-functional: Collaboration enhancement



A stylized, painterly illustration of a mountain range. The mountains are rendered in soft, pastel shades of blue, pink, and yellow, with a dreamy, ethereal quality. A bright, four-pointed star or spark is positioned at the top center, directly above the highest peak. The overall composition is centered and balanced, with the text 'Thank You' prominently displayed in the middle.

Thank You